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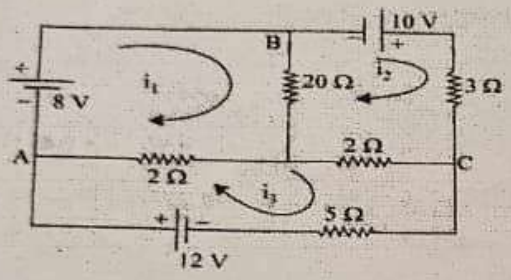
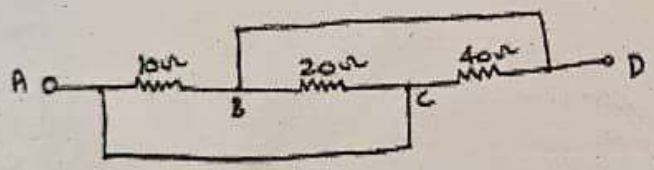
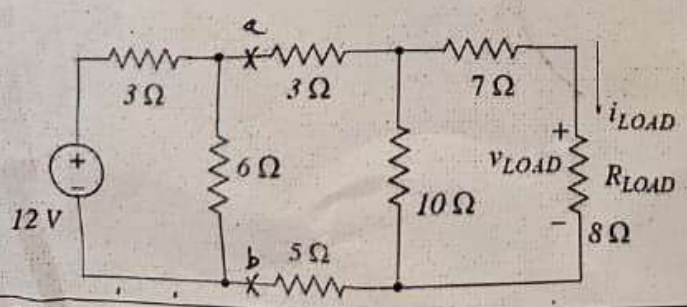
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Tel: 080-68188192

**Department of Electrical and Electronics**

Basics of Electrical Engineering

| | | | |
|---|---|---|---|
| **Date** | **13th May 2025** | **Maximum Marks** | **50** |
| **Course Code** | **EE123ATD** | **Duration** | **90 Mins** |
| **Sem** | **II Semester** | **Test-1** | |
| **Basics of Electrical Engineering** | | | |

| Q.No | Part B Test-Questions | Marks | COs | BT |
|---|---|---|---|---|
| 1a) | State KCL and KVL Determine current in 5-Ω resistor for the circuit shown below.   | 6 | 02 | 02 |
| b) | Define ohm's law. Mention the limitations of Ohm's Law. Calculate the effective resistance $R_{AD}$ in the network shown.   | 4 | 01 | 02 |
| 2a | Using Thevenin's Theorem find $V_{TH}$, $R_{TH}$ and the load current $I_L$ flowing through and load voltage across the load resistor in the circuit below.   | 6 | 03 | 03 |
| b | State and prove the maximum power transfer theorem. | 4 | 01 | 01 |
| 3a | Define following terms 1) Frequency 2) Instantaneous value 3) Time period 4) Power factor 5)Angular velocity. | 5 | 01 | 01 |

| | | | |
|---|---|---|---|
| b | An alternating current of 50 Hz has a maximum value of 110 A. Find (i) write the equation for the instantaneous value of the current. (ii) the time taken to reach 80 A for the second time. | 5 | 02 |
| 4a) | A series RL circuit takes 384 watts at a p.f of 0.8 from 120 V, 60Hz supply. What are the values of R and L. | 4 | 0 |
| b) | Analyze the phase relationship between voltage and current in an RC circuit with a phasor diagram. | 6 | 0 |
| 5a) | A current of 5A flows through a non-inductive resistance in series with a choke coil when supplied at 250 V, 50 Hz. If the voltage across the resistance is 125 V and across the coil 200 V, calculate : (i) Impedance, reactance and resistance of the coil,(ii) The power absorbed by the coil, (iii) The total power. Draw the vector diagram. | 6 | ( |
| b | A current of 10A flows in a series circuit consisting of R=10 ohm, L=0.1H and C=150 micro farad. Find the power and impedance, if the frequency of the supply is 50 Hz. | 4 | |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| Marks Distribution | Particulars | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Test | Max Marks | 13 | 25 | 12 | - | 4 | 34 | 12 | - | - |

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Department of Electrical and Electonics

# DEPARTMENT OF
# ELECTRICAL AND ELECTRONICS ENGINEERING

| | | | |
|---|---|---|---|
| Date | 17th June. 2025 | Maximum Marks | 50 |
| Course Code | EE123AT | Duration | 90 Min. |
| Sem | II Semester | CIE-2 | |
| **BASICS OF ELECTRICAL ENGINEERING** | | | |

| Question. No. | QUESTIONS | Marks | BT | CO |
|---|---|---|---|---|
| 1 a. | Write down advantages of three-phase AC system single-phase AC systems. | 04 | 2 | 1 |
| 1 b. | A 600 kVA single -phase transformer has an efficiency of 92% both at full-load and half -load at unity power factor. Determine its efficiency at 75% of full load at 0.9 power factor lag. | 06 | 3 | 2 |
| | | | | |
| 2 a. | With a neat circuit and vector diagram prove that two wattmeter is enough to measure three phase power. | 06 | 3 | 1 |
| 2 b. | A single phase, 20kVA transformer, has 1000 primary turns and 2500 secondary turns. The net cross-sectional area of the core is 100 cm2. When the primary winding is connected to 500V, 50Hz supply, calculate<br>i) The maximum value of the flux density in the core<br>ii) The voltage induced in the secondary winding and<br>iii) The primary and secondary full-load currents | 04 | 3 | 2 |
| | | | | |
| 3 a. | With neat sketch explain the constructional features of a core type and shell type transformers. | 06 | 2 | 1 |
| 3 b. | A 3-phase, 400V, induction motor takes an input of 40kW at 0.45 pf lagging. Find the reading of each of the two single phase wattmeter's connected to measure the input. | 04 | 2 | 1 |
| | | | | |
| 4 a. | Derive the EMF equation of a transformer from fundamentals. | 04 | 3 | 2 |
| 4. b | A balanced 3- phase, star connected load of 210kW load takes a leading current of 160 A at 1100V, 50 Hz. Find the parameters per phase. | 06 | 3 | 2 |

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (EVEN SEM-UG)

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Department of Electrical and Electonics

| | | | | |
|---|---|---|---|---|
| 5 a. | Justify the following statements:<br>a) In a transformer core losses are constant whereas copper losses are variable.<br>b) The delta-connected load consumes three times more power than if it were connected in a star configuration. | 06 | 4 | 3 |
| 5 b. | The required No-load ration in a 125kVA, 50 Hz, single phase core – type transformer is 6600 / 150 V. Find the number of turns per limb on the high and low voltage sides , if the flux is to be about 0.059 Wb. Also calculate the currents in both the windings | 04 | 3 | 2 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| Marks Distribution | Particulars | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Test | Max Marks | 20 | 24 | 06 | ---- | ---- | 14 | 30 | 06 | ---- | ---- |

Department of Electrical & Electronics Engineering

**Academic Year 2024-25 (EVEN SEM-UG)**

| Date | 09 Mar 2025 | Maximum Marks | 50 |
|---|---|---|---|
| Course Code | EE123ATD | Duration | 90 Mins |
| Sem | II | IMPROVEMENT CIE | |
| Basics of Electrical Engineering | | | |

| Q.No | Test Questions | Marks | COs | BT |
|---|---|---|---|---|
| 1. | (a) With a neat sketch and vector diagram, mathematically explain how the rotating magnetic field is produced in the three phase induction motor | 06 | 02 | 02 |
| | (b) A 4 pole 50 Hz induction motor has a slip of 1% at NO load. When operated at full load, the slip is 2.5%. Find the change in speed from no load to full load. | 04 | 03 | 03 |
| 2. | (a) What are the two types of three phase induction motors. Distinguish its constructional features | 05 | 02 | 02 |
| | (b) List the different types of single-phase induction motor. Explain principle of operation for any one type. | 05 | 02 | 02 |
| 3. | (a) Draw the torque-slip characteristic curve of a three-phase induction motor. Clearly mark and label the following regions on the curve:<br>(a) Negative slip region; (b) Zero slip; (c) Maximum torque point;<br>(d) Positive slip region;<br>Indicate its significance. | 05 | 04 | 04 |
| | (b) List any five precautions against the electric shock. | 05 | 04 | 02 |
| 4. | (a)Describe the working of miniature circuit breaker (MCB) | 04 | 04 | 02 |
| | (b) What is necessary of earthing? With a neat sketch, explain plate earthing. | 06 | 04 | 02 |
| 5. | (a) Draw the single line diagram of power transmission and distribution mentioning the different voltage levels. | 03 | 01 | 03 |
| | (b) Estimate the total daily energy requirement for the loads shown in the below table. Also calculate monthly electricity bill if the electric cost is Rs.2/unit for each 40 units, Rs.4/unit for next 50 units and Rs.6/unit for rest of the units. | 07 | 04 | 04 |

| *Name of the appliance* | *Power Rating (W)* | *Average daily usage hrs.* | *Number of Hours* |
|---|---|---|---|
| *Fan* | *50* ×4 | *10* | *4* |
| *CFL* | *12* ×2 | *6* | *8* |
| *LED TV* | *100* ×6 | *5* | *2* |
| *Refrigerator* | *300* ×8 | *24* | *1* |
| *Laptop* | *60* ×4 | *5* | *1* |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| Marks Distribution | Particulars | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Test | Max Marks | - | 16 | 04 | 30 | - | 26 | 08 | 16 | - |

**********

EE113ATD

USN | 1 | R | V | 2 | 4 | C | H | 0 | 0 | 4 |

# RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I Semester B. E. Regular / Supplementary Examinations Feb/Mar-2025

## Electrical and Electronics Engineering

## BASICS OF ELECTRICAL ENGINEERING

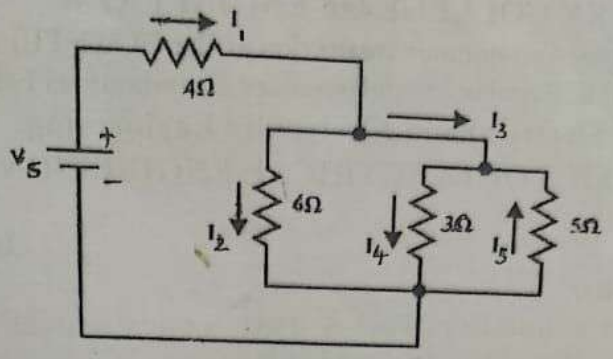
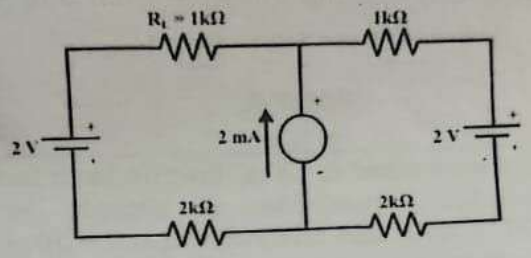
*Time: 03 Hours* *Maximum Marks: 100*

*Instructions to candidates:*

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 is compulsory. Choose any one full question from 3 or 4, 5 or 6, 7 or 8 and 9 or 10.

**PART-A**

| | | | M | BT | CO |
|---|---|---|---|---|---|
| 1 | 1.1 | The peak factor of a sine wave is 1.414, find its form factor. | 02 | 1 | 2 |
| | 1.2 | A sinusoidal voltage wave in a circuit is given by $e = 100\sin(628t + 30^0)$ *volts* and the current is $i = 10\sin(628t - 60^0)$, the power factor and the power in the circuit is _____. | 02 | 1 | 2 |
| | 1.3 | A $100W$ bulb is connected across a $200V$ mains, what is the resistance of the bulb, the current drawn and if another $100W$ bulb is added in parallel with it, what will be the total current supplied from the mains? | 02 | 1 | 1 |
| | 1.4 | An impedance of $Z_1 = 10\angle 30^0\Omega$ is connected in series with another impedance of $Z_2 = 15\angle 60^0\Omega$ . The total impedance is _____ and the power factor of the circuit is _____. | 02 | 2 | 2 |
| | 1.5 | A 3–phase circuit is drawing current of $12A$ from a $400V$ 3-phase supply. If the circuit is a star connected resistors, then the impedance per phase and the total power drawn is _____. | 02 | 2 | 2 |
| | 1.6 | What is the importance of voltage regulation in transformers? | 02 | 1 | 3 |
| | 1.7 | A 3-phase, 4-pole, $50Hz$ induction motor is running at a slip of 4%. What is the running speed of the motor? | 02 | 1 | 2 |
| | 1.8 | List the differences between cage type and slip ring type induction motors. | 02 | 2 | 3 |
| | 1.9 | What is the purpose of 'fuse' and 'miniature circuit breaker' which is the most preferred one between the two. | 02 | 3 | 1 |
| | 1.10 | What is earthing and why it is necessary? | 02 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | State and explain the maximum power transfer theorem. | 04 | 2 | 1 |
| | b | In the circuit shown in Fig 2b, determine $I_1, I_2$ and $I_3$ when $I_4$ is $4A$ and also find the input voltage $V_S$.   Fig 2b | 06 | 3 | 2 |
| | c | Find the current flowing through $R_L$ in the circuit shown in Fig 2c.   Fig 2c | 06 | 3 | 2 |
| 3 | a | Show that the power expression of a single phase circuit is $P = VI\cos\phi$. | 06 | 2 | 1 |
| | b | Explain what you mean by phase and phase difference? | 04 | 2 | 1 |
| | c | A voltage of $120V$ at $50Hz$ is applied to a resistance $R$ in series with a capacitance $C$. The current drawn is $2A$ and the power loss in the resistance is $100W$. Calculate the resistance and the capacitance and the power factor. | 06 | 3 | 2 |
| | | **OR** | | | |
| 4 | a | List the properties of a series resonant circuit. | 04 | 1 | 1 |
| | b | Draw the vector diagrams of a series $R$-$L$, $R$-$C$ and $R$-$L$-$C$ circuits. | 06 | 2 | 1 |
| | c | Two emf waveforms are given by $e_1 = 200\sin\omega t$, $e_2 = 200\sin(\omega t + 30)$, find the phase angle between them when they are in series. What is the total rms voltage? When the total is applied across a resistor of 8Ω, what will be the current and power drawn by the resistor? | 06 | 3 | 2 |
| 5 | a | What do you mean by balanced three phase vectors? Write the representation of balanced three phase voltages in<br>i) Vector diagram<br>ii) Wave forms<br>iii) Instantaneous values on sequence 1, 2, 3. | 06 | 2 | |
| | b | A balance load of $(9 - j6)\Omega$ is connected in delta across a balanced 3-phase source of $100V$. Find the phase and the line currents and the active, reactive and apparent powers and also the power factor. | 05 | 3 | |

| | | | | | |
|---|---|---|---|---|---|
| | c | Show that the two watt meters are sufficient to measure power in 3-phase circuits. | 05 | 2 | 2 |
| | | **OR** | | | |
| 6 | a | Discuss the working principle and types of single phase transformers. | 05 | 1 | 1 |
| | b | Define the various losses of a transformer. Discuss how losses vary. Also mention the condition for maximum efficiency. | 05 | 2 | 2 |
| | c | The efficiency of a $400kVA$, single phase transformer is 99.77% when delivering full load at 0.8 power factor and 99.13% at half full load and unity power factor. Calculate the iron loss and the full load copper loss. | 06 | 3 | 3 |
| | | | | | |
| 7 | a | Explain the process of torque production in 3-phase induction motors. | 06 | 2 | 2 |
| | b | Bring out the comparison between the two types of 3-phase induction motors. | 05 | 2 | 2 |
| | c | Explain why the starting torque of a single phase induction motor is zero. | 05 | 2 | 3 |
| | | **OR** | | | |
| 8 | a | Discuss the different methods of starting single phase induction motors and mention the applications of a single phase induction motors. | 06 | 2 | 2 |
| | b | Draw the typical torque slip characteristics of a 3 phase induction motor and mark all the salient points on this curve. | 06 | 2 | 3 |
| | c | If the emf in the stator of an 8-pole induction motor has a frequency of $50Hz$ and that of the rotor is $1.5Hz$, at what speed is the motor running and what is the slip? | 04 | 3 | 3 |
| | | | | | |
| 9 | a | Explain the architecture of electrical power generation, transmission and distribution through a relevant block diagram. | 06 | 2 | 1 |
| | b | Give a detailed comparison between a fuse and an $MCB$. | 05 | 2 | 2 |
| | c | What do you mean by earthing? Why is it necessary and what is its importance? | 05 | 2 | 2 |
| | | **OR** | | | |
| 10 | a | A domestic house has the following loads:<br>i) Geysers 2 numbers, each $2kW$, used for 4hours per day<br>ii) $LED$ tube lights 5 numbers, each $22W$, used for 6hours per day<br>iii) Electric stoves 2 numbers, each $2kW$, used for 4hours a day<br>iv) Ceiling fans 6 numbers, each $55W$, used for 8hours a day<br>v) Pump 1 number, of 1hp used for 2hours a day<br>Calculate the monthly energy consumption and the bill amount if the rate is $Rs.4.70$ for first 100units and $Rs.7.00$ for the subsequent units. The tax on the amount is 2.5% and there is a rebate on the solar unit equipped in this house of $Rs.50$. | 07 | 4 | 4 |
| | b | Mention the dos and don'ts to avoid electric shocks. | 04 | 2 | 4 |
| | c | Explain the plate earthing with diagram, which one is preferred much. | 05 | 2 | |